COURSE SYLLABUS
ECON 4760/6760: *Time Series Analysis*
Spring 2024

Terry College of Business
John Munro Godfrey, Sr. Department of Economics
UNIVERSITY OF GEORGIA

Instructor:
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Office hours: Tuesday 8:30a-10:45a

Course details:
MWF 9:10a-10:00a | 116 Correll Hall
Exam 1: Feb 26 | Exam 2: Apr 26 | Data project due date: May 6, 11a
Arkaive enrollment code: Z826

Course description:
This course is an introduction to the econometric analysis of time series data. Unlike most cross-sectional data, the sequential ordering of time series data contains useful information — time series observations typically exhibit *serial dependence*. The focus of the class is on how models of time-dependent data can be used to forecast and make causal inference about economic behavior. Students should know the basics of probability and statistical inference, understand the techniques and uses of linear regression models, and be comfortable with linear algebra.

Course objectives:
After taking this course you will
1. understand the fundamental difference between time series and cross section data;
2. construct and interpret time series graphs;
3. understand the nature of stationary stochastic processes;
4. estimate and interpret distributed lag models;
5. estimate and interpret ARIMA models;
6. identify causal effects in vector models; and
7. understand the nature of non-stationarity in vector models.

Topic outline (chapter in lecture notes):
1. Nature of time series (2)
2. Time series fundamentals (3)
3. Dynamic regressions (4)
4. ARIMA models (5)
5. Vector (multivariate) processes (6)
6. Causaility in VAR models (7)
7. Cointegration (8)

Reading:
- Required: Lecture notes available on course page. We will not cover all the material in the lecture notes; you will be responsible only for the sections of the notes that correspond to class lectures.

- Supplemental: Rob J. Hyndman and George Athanasopoulos, *Forecasting: Principles and Practice*, 3rd edition. This is a free text that is potentially useful as a supplemental resource for some of the topics we will cover. It includes extensive time series coding examples in **R**.

Software:
The course entails some hands-on data analysis using the tools of time series. For this purpose you will need access to statistical software that can work with time series data. I encourage using **R**, since it is free and most of you will have some experience with it, but **Python** and **Stata** are acceptable. While I will discuss some coding examples in class, primarily if not exclusively in **R**, I am not able to provide detailed coding instruction in or out of class. For **R**, I recommend the free text **R** for Data Science for help on the basics of **R** and the examples from the supplemental text above. The Terry College Analytics Lab is available to provide programming assistance, primarily with **R** coding. I encourage you to use their services as needed (contact: Taylor Gorak, taylor.gorak@uga.edu).

Grading:
The final course grade depends on your performance on two in-class term exams, approximately six homework assignments, a data analysis project (which counts as the final exam) and an attendance score, according to these weights:

Exam average	40%
Homework assignment average	30%
Data project	20%
Attendance score	10%

The schedule for the in-class term exams is given above, as is the deadline for the data project. Exams will be given during the usual class period on the scheduled day. There will be no make-up exams. Term exams are 'open-book' — you can have access to and use any course material during the test. You are not, however, allowed to confer with anyone inside or outside the classroom during the test. Homework assignments and the data project will be available through the Assignments section of the course home page. Homework and projects submitted after the deadlines will be given a grade of zero. I assign course letter grades, with plus-minus distinctions, based on my assessment of your ranking in the final course grade distribution. There is no fixed grading scale, but you should expect that the minimum course grade to pass the course with a 'C' grade is 60.

Attendance score:
Your baseline attendance score is 100. Unexcused absences will reduce this baseline. You can miss up to three classes without penalty. Each absence beyond this limit will reduce your attendance score by 15 points. For example, three absences during the semester imply an attendance score of 100, four absences imply a score of 85, five a score of 70, and so on. Class participation — asking questions, answering questions, making comments, etc. — is strongly encouraged and, at my judgment, can increase your attendance score.

I monitor attendance using Arkaive. Download the Arkaive app on your phone or tablet and follow the instructions to sign-up using the code at the top of this syllabus. You are responsible for checking-in to class each session using the app on your device. I have set up Arkaive to allow check-in ten minutes before the scheduled start of the class through fifteen

minutes after the start (a twenty-five minute check-in window). You will not be able to check-in when the window is closed. The app will recognize your device only if you are physically present in the classroom. I will consider excusing absences (beyond the 3-class limit) only for documented illness or official school-related functions. Make excusal requests through Arkaive. You do not need to sign into Arkaive on exam days.

Electronic devices:
You are allowed to use laptops and other electronic devices during class, but only for course-related purposes. Phones and other devices should be turned off or muted during class.

Academic honesty:
I expect all students in this course to fully understand and comply with UGA's culture of academic honesty: *As a University of Georgia student, you have agreed to abide by the University's academic honesty policy, "A Culture of Honesty," and the Student Honor Code. All academic work must meet the standards described in "A Culture of Honesty" found at: www.uga.edu/honesty. Lack of knowledge of the academic honesty policy is not a reasonable explanation for a violation. Questions related to course assignments and the academic honesty policy should be directed to the instructor.*

UGA Well-being Resources:
UGA Well-being Resources promote student success by cultivating a culture that supports a more active, healthy, and engaged student community. Anyone needing assistance is encouraged to contact Student Care & Outreach (SCO) in the Division of Student Affairs at 706-542-8479 or visit sco.uga.edu. Student Care & Outreach helps students navigate difficult circumstances by connecting them with the most appropriate resources or services. They also administer the Embark@UGA program which supports students experiencing, or who have experienced, homelessness, foster care, or housing insecurity.

UGA provides both clinical and non-clinical options to support student well-being and mental health, any time, any place. Whether on campus, or studying from home or abroad, UGA Well-being Resources are here to help.

- Well-being Resources: well-being.uga.edu
- Student Care and Outreach: sco.uga.edu
- University Health Center: healthcenter.uga.edu
- Counseling and Psychiatric Services: caps.uga.edu or CAPS 24/7 crisis support at 706-542-2273
- Health Promotion/ Fontaine Center: healthpromotion.uga.edu
- Disability Resource Center and Testing Services: drc.uga.edu

Additional information, including free digital well-being resources, can be accessed through the UGA app or by visiting https://well-being.uga.edu.

This syllabus is a general plan for the course; deviations announced to the class by the instructor may be necessary.